

Vig — one-page write-up

It never takes a position it can't cover. Paper account `PA3PLC0VLR4I` · github.com/emmanuelist/vig · emmanuelist.github.io/vig

AI LOGIC

Vig sells **iron condors** on SPY, QQQ and IWM — a short put spread below the market and a short call spread above it, collecting premium from both sides. Three constraints converge on that instrument: Alpaca's `--legs` accepts at most four and a condor is exactly four; only one side can ever be breached, so maximum loss is `width - credit` rather than twice it, making the structure self-covering; and it is delta-neutral, the honest posture when the window is too short for a directional edge to be statistically real.

Selection is mechanical. Short strikes sit at ≈ 0.12 delta from the live chain, wings one width out, with width scaled as a fraction of spot rather than a fixed dollar amount — a flat \$5 is 0.66% of SPY but 1.7% of IWM and would silently exclude the cheaper underlying. Credit is priced **against us**, selling at the bid and buying at the ask, so the reserve is what we would collect if every fill went the wrong way.

The one judgement that is not arithmetic goes to a language model. Whether *today* is a bad day to be short premium lives in prose, so Vig pulls headlines via `alpaca data news` and asks a model on Featherless AI whether an identifiable event makes a range-bound assumption unsafe. It can only ever **veto** — never increase size or loosen a limit — and if it is unavailable the agent keeps trading and says so on screen, because a dead endpoint must not resemble approval.

RISK GATES

No structure reaches the broker without clearing all eight.

GATE	REFUSES WHEN
<code>illiquid</code>	any leg's bid/ask exceeds \$0.15
<code>thin-credit</code>	credit is under 10% of width
<code>position-too-large</code>	one structure risks over 5% of equity
<code>bucket-full</code>	one underlying+expiry exceeds 6% of equity
<code>book-full</code>	total reserved exceeds the 25% ceiling
<code>no-wing</code>	the protective strike does not exist
<code>inverted</code>	short strikes crossed; loss no longer bounded
<code>regime</code>	the newswire says stand aside

The per-bucket cap exists because the total ceiling is otherwise a fiction: twenty identical SPY condors are not twenty bets, they breach together as one position at twenty times the size. Refusals carry readable reasons and are shown beside the fills.

The account finished down 13%. On 2 September at 23:48 a `position list` call failed; the tick substituted an empty array, which the reconciler could not tell from an account holding nothing. It marked every structure settled and reset reserved to zero — `book-full` then saw a full ceiling against a book already full, and the agent doubled its position count before the market rallied through the short calls. Every structure stayed inside its own bounded loss and **uncovered exposure never left zero**. The per-position guarantee held; the portfolio ceiling did not, defeated by a bug rather than a decision. The failure and its two-guard fix are both in the repository history.

ALPACA INFRASTRUCTURE

The CLI is the entire execution path — `data option chain` for selection, `order submit --order-class mleg` for the four-leg structure, plus `position list`, `account get`, `clock` and `data news`. Not an SDK with the CLI mentioned in passing: **4,594 invocations**, each leaving a receipt with its argv, exit code and duration.

Every order is submitted twice — once with `--dry-run`, which renders the exact request without sending it, then again as the identical argv for real. The proof and the order are the same request: 48 submissions, 24 dry-run proofs, one per structure, committed under `evidence/`.

Two chain behaviours are defended against, both found the expensive way: the chain returns **already-expired contracts** (470 dead SPY strikes on 2 September), and `greeks` appears only on unexpired ones, making a zero delta the only reliable liveness signal. Index options were evaluated when Alpaca launched them mid-event and declined on measurement — 400 of 400 SPX contracts carried a bid and none carried a delta, and selection here is delta-based.

Paper trading only. `ALPACA_LIVE_TRADE` is never set.